

Laptop Price Prediction — Documentation

1. Project Overview

Objective: Build a machine learning model to predict laptop prices from product specifications. This document captures every step performed in the project, the exact numerical results, and the reasoning behind choosing Linear Regression as the baseline model.

2. Environment & Dependencies

The project was implemented in Python using standard data science libraries: pandas, numpy, matplotlib, seaborn, and scikit-learn.

3. Dataset: Source & Structure

The dataset contains rows of laptop specifications and their corresponding retail prices. Key points:

- Target variable (y): 'Price'
- Features (X): categorical and numerical features such as brand, processor, RAM, storage, etc.
- Train/Test split used: 80% training, 20% testing (random_state=42).

4. Exploratory Data Analysis (EDA)

- Inspected dataset shape and sample values.
- Checked datatypes and missing values.
- Generated descriptive statistics for numerical features.
- Visualized distributions of numerical features and target variable.
- Created bar plots for categorical columns.
- Generated correlation matrix for numerical features.
- Used scatter plots to visualize feature-target relationships.

EDA helped identify missing values, outliers, and confirmed linear relationships for several features.

5. Data Preprocessing

Steps applied to prepare data for modeling: 1) Handled missing values (imputation or row removal). 2) Removed duplicate rows. 3) Converted datatypes where required. 4) Encoded categorical variables using one-hot encoding. 5) Split the dataset into training and testing subsets. 6) Feature scaling considered optional for linear regression.

6. Model Selection — Why Linear Regression?

Linear Regression was chosen for the following reasons: - Simplicity and interpretability: feature coefficients explain impact on price. - Baseline benchmark: useful for comparing with more complex models. - Fast and computationally inexpensive. - Suitable when exploratory analysis suggests linear relationships. Assumptions (linearity, independence, equal error variance, normality of residuals) were considered acceptable for initial modeling.

7. Model Evaluation — Metrics & Interpretation

The model was evaluated using the following metrics: - Mean Squared Error (MSE): measures average squared prediction error. - R^2 Score: measures proportion of variance in target explained by the model. Approximate Accuracy: derived from Mean Absolute Percentage Error.

Metric	Value	Interpretation
Mean Squared Error (MSE)	576,410,115.15	Large due to scale of target variable.
R^2 Score	0.7042	Explains about 70% of variance in price.
Approx. Accuracy	81.84%	Model achieves ~82% average accuracy.

8. Test Cases — Sample Outputs

Actual Price	Predicted Price	Abs Error	Absolute % Error
33,690	33,809.08	119.08	0.35%
86,990	72,695.55	14,294.45	16.43%
39,490	41,486.95	1,996.95	5.06%
63,990	79,625.64	15,635.64	24.43%
134,990	139,332.94	4,342.94	3.22%

Interpretation: - Predictions for some laptops are very accurate (<5% error). - Larger errors occur in cases where nonlinear effects or unrepresented specifications exist. - Overall, the model generalizes reasonably well with ~82% average accuracy.